

Data Processing

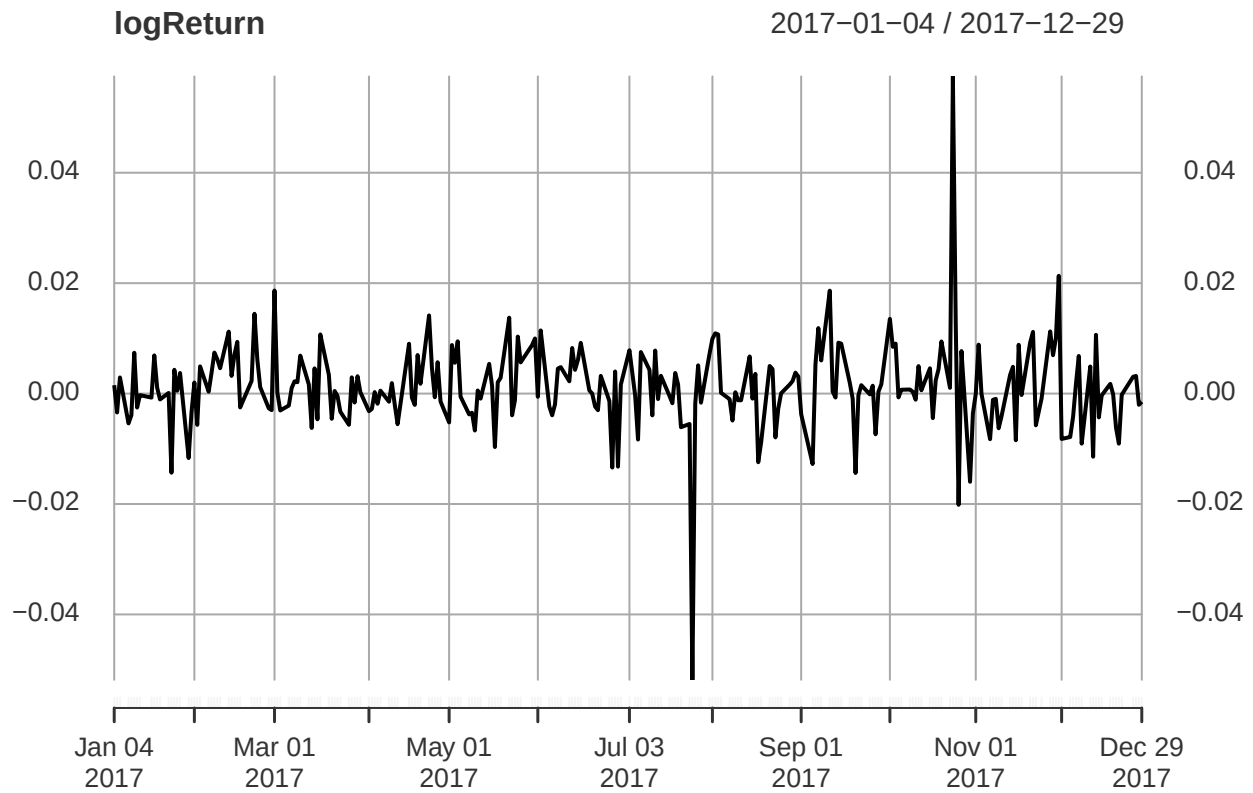
```
library('quantmod')

## Loading required package: xts
## Loading required package: zoo
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric
## Loading required package: TTR
## Version 0.4-0 included new data defaults. See ?getSymbols.
adj = getSymbols("MMM", from="2017-1-1", to="2017-12-31", auto.assign=F)[,6]

## 'getSymbols' currently uses auto.assign=TRUE by default, but will
## use auto.assign=FALSE in 0.5-0. You will still be able to use
## 'loadSymbols' to automatically load data. getOption("getSymbols.env")
## and getOption("getSymbols.auto.assign") will still be checked for
## alternate defaults.
##
## This message is shown once per session and may be disabled by setting
## options("getSymbols.warning4.0"=FALSE). See ?getSymbols for details.
##
## WARNING: There have been significant changes to Yahoo Finance data.
## Please see the Warning section of '?getSymbols.yahoo' for details.
##
## This message is shown once per session and may be disabled by setting
## options("getSymbols.yahoo.warning"=FALSE).

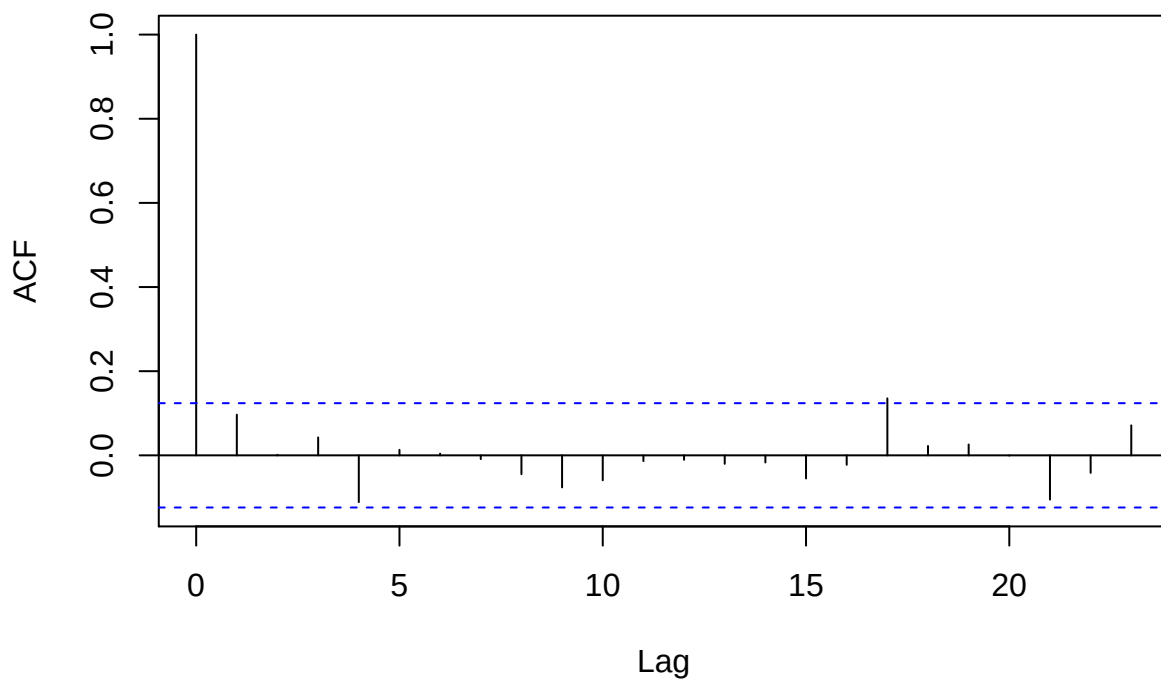
Problem 4

#(a)
logReturn = diff(log(adj))[-1]
plot(logReturn)
```



```
#(b)  
acf(logReturn)
```

Series logReturn



```
##(c)
Box.test(logReturn, lag = 20, type = "Ljung-Box")

##
## Box-Ljung test
##
## data: logReturn
## X-squared = 15.52, df = 20, p-value = 0.7459
p-value of 0.7459 suggests significant evidence of correlation.
```